

Structure-Aware Resistance Prediction in *Mycobacterium tuberculosis*

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1. Abstract

This project explored multiple modeling strategies to predict drug resistance, beginning with classification approaches based on WHO confidence grades using methods such as logistic regression and gradient boosting. The focus then shifted to a regression framework using positive predictive value (PPV) as a more informative and continuous target. We evaluated different feature representations, including molecular fingerprints and learned embeddings (e.g., ChemBERTa), and incorporated additional data sources such as the CRyPTIC dataset. Further improvements were achieved through genomic position-based feature engineering, capturing the biologically relevant context of mutations. Across these stages, each iteration of the model demonstrated incremental performance gains, highlighting the value of progressively integrating richer representations and domain-specific information.

2. Introduction

2.1. Historical Context

Mycobacterium tuberculosis was first discovered by Dr. Robert Koch on March 24, 1882. However, the bacteria themselves are estimated to have been present for the past 3 million years [1]. An infection from *M. tuberculosis* causes Tuberculosis (TB) disease in individuals. Active TB disease is mainly characterized by coughing for three weeks or longer, chest pain, and coughing up blood or sputum, and infection is verified through TB skin or blood testing [2]. Even if an infected individual does not display symptoms, they are still at risk of spreading the disease. Since TB is an airborne disease and infected individuals do not always have symptoms, it remains the deadliest infectious disease [2].

2.2. The Presented Problem

Current treatment plans for TB disease include various combinations of antibiotics, with the top six being: Ethambutol, Isoniazid, Moxifloxacin, Rifampin, Rifapentine, and Pyrazinamide [3]. It is important that individuals on a TB treatment plan take all the medication prescribed to them even when symptoms alleviate [3]. The main risk for not completing treatment is reinfection, which can potentially lead to developing drug-resistant TB [3]. Treating drug-resistant TB is a complicated process that requires one to visit a TB expert and undergo an extensive treatment plan [4].

As TB remains a leading cause of infectious disease mortality worldwide [5], along with multidrug-resistant and extensively drug-resistant strains, we are presented with an escalating threat to global public health. Phenotypic drug susceptibility testing, while considered the gold standard, requires weeks [6] to yield results due to the slow growth of *M. tuberculosis*, delaying the initiation of effective treatment. Genotypic approaches based on whole-genome sequencing offer a rapid alternative by identifying mutations associated with drug resistance, but current

interpretation frameworks [7] function as static lookup tables that treat each drug as a categorical label, without leveraging the structural chemistry of the drugs themselves.

2.3. Our Project Objective

The initial objective of this project was to develop a model that predicts antimicrobial resistance by jointly encoding a genetic mutation and a molecular representation of the drug, enabling the model to learn relationships between mutational context and drug chemistry and potentially generalize to novel anti-TB compounds or novel mutations.

We chose to explore the WHO’s “Catalogue of Mutations in *M. tuberculosis* Complex and Their Association with Drug Resistance” [8] [9]. The WHO’s 2024 mutation catalogue provides a large, high-quality dataset linking *Mycobacterium tuberculosis* mutations to antibiotic resistance phenotypes, combining whole-genome sequencing with validated drug susceptibility results. This dataset included over 48,000 clinical aggregates with curated features capturing tens of thousands of genetic variants across key resistance genes, along with resistance profiles for 15 anti-tuberculosis drugs, which enables our goal of jointly modeling mutation profiles and drug molecular structure, supporting a structure-aware approach to learning mutation–drug interactions. However, a major challenge in the WHO dataset was the severe class imbalance and the lack of meaningful interpretation of mutation location.

Based on the findings from our EDA (**Section 4.1**), the objective for this project went from predicting if bacteria will be resistant to a new drug based on prelearned mutation and drug structure, to predicting how resistant bacteria will be to the existing TB drugs given a new mutation. We concluded that we would want mutations and antibiotic drug structures as our main features to train our model, and predictive antibiotic resistance values as our target.

3. Methods

3.1. Preprocessing

Over 70% of samples are labeled “Uncertain significance,” while the clinically actionable classes (“Associated with Resistance” and “Not associated with Resistance”) make up less than 2% combined. Without proper handling, a model would default to predicting the majority class, failing to learn meaningful mutation–drug relationships. Addressing this imbalance is critical to capturing biologically relevant patterns, and to do so, we designed a preprocessing and modeling pipeline that emphasizes both balance and multimodal representation.

First, we applied strategies such as balanced sampling, class weighting, and evaluation metrics like precision and recall to ensure the model focuses on clinically meaningful predictions rather than overall accuracy. We also carefully considered how to handle ambiguous labels, since these “uncertain” cases reflect real biological uncertainty that may influence how the model learns mutation and drug relationships. Ultimately, to address the class imbalances, we removed uncertain resistance values when creating our final dataset to ensure our models were trained only on meaningful and well-balanced data.

Crucially, to support our structure-aware framework, we integrated both genetic and chemical information. Mutation data was custom encoded into machine-learning-compatible features using a parser to break down protein and RNA/DNA level mutations into their associated mutation types (e.g., frameshift, deletion, nonsense, duplication). We used Mycobrowser [10] to facilitate genomic structure embeddings of the mutations. This allows our model to understand the importance of the positioning of the mutation(s) and their relationship to resistance.

Drug information was represented using molecular descriptors such as SMILES strings to a Morgan fingerprint, with each fingerprint bit encoded into its own column. Morgan fingerprints represent specific substructures, including rings and functional groups, using binary encoding, allowing for very precise structural patterns. ChemBERTa embeddings, open source BERT-style transformers trained on chemical datasets including PubChem, ZINC20, and MoleculeNet [11], were also used to capture each drug’s global structure using higher-dimensional vector representations.

Finally, supplemental minimum inhibitory concentration (MIC) measurements were added from the CRyPTIC Consortium, which included over 12,000 isolated samples of *M. tuberculosis*, with over half of the samples having confirmed resistance to TB drugs [12]. Similar to the WHO dataset, this dataset included genetic variants and their associated resistance to the 15 main TB drugs.

Using higher-level molecular and mutation encodings will provide a unified feature space where the model can learn interactions between mutation profiles and drug structure, directly supporting our goal of generalizing resistance values to drugs given a mutation.

3.2. Exploratory Data Analysis

Our dataset initially contained 114 features before preprocessing. With the addition of custom encoding, our new dataset expanded to over 2000 features. When trying to understand the relationship between features in our dataset and which features we want to train our model with, we chose a select number of features that could have a significant explanation for what contributes to drug resistance. Our initial choice was mutation type, mutation location (gene), associated drug, and resistance grade for our correlation heatmap. We then applied dimensionality reduction techniques like PCA and UMAP to better understand the structure of this high-dimensional, multimodal feature space before modeling.

Several UMAP projections were run to better understand what could explain the resistance grade. Our initial projection included all of our features, with our next two projections looking specifically at mutation types and drug structures. Our target feature for all projections was the WHO’s specified resistance grade.

Another question that arose during our EDA process regarded the structural similarities between our drugs. Drugs with similar structures could severely impact the accuracy of the model. We used Tanimoto coefficients on Morgan fingerprints to quantify structural similarity between drugs, where higher values indicate greater overlap in chemical substructures. The

Tanimoto Theory formula (**Figure 1a**) helps to look at the intersecting features of two drugs over the total unique features in each drug. It can be used for Morgan fingerprints (**Figure 1b**) to give us a coefficient score. With this, we could then hide a subset of drugs with similar structures, which then motivated evaluating model generalization by holding out structurally similar drugs, testing whether learned chemical patterns can transfer to unseen or less-characterized compounds. A threshold of ≥ 0.7 was set for the coefficients to act as our similarity threshold.

$$T_c(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Figure 1a. Tanimoto Theory formula

$$T_c = \frac{c}{a + b - c}$$

Figure 1b. Tanimoto equation for Morgan fingerprints.
c: Number of bits set to (1) in both fingerprints (common features).

a: Number of bits set to (1) in fingerprint A.

b: Number of bits set to (1) in fingerprint B.

$A + b - c$: Total number of unique "on" bits (union)

3.2.1. EDA Results

3.2.1.1. Correlation Heatmap

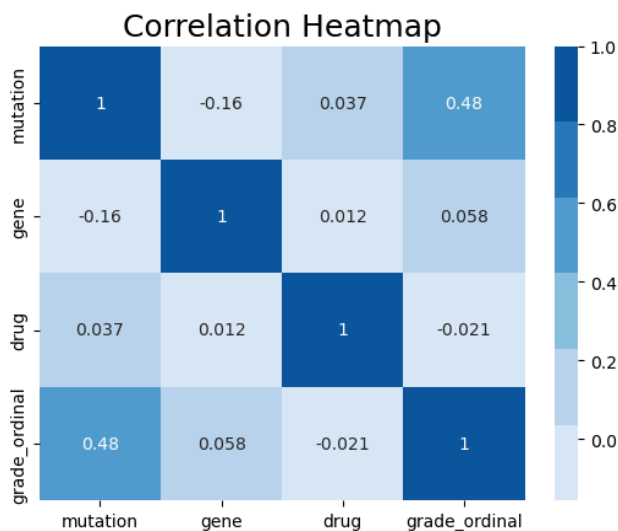


Figure 2a. Initial understanding of the correlation between select features: what mutation occurred (mutation), which gene the mutation occurred in (gene), their associated drug (drug), and their resistance scoring (grade_oridal).

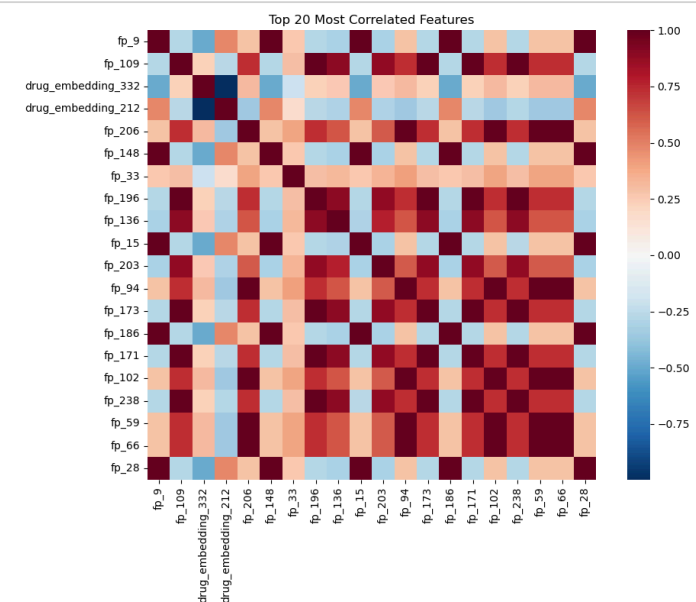


Figure 2b. Correlation heatmap of the 20 most correlated features with the CRyPTIC dataset and ChemBERTa embeddings

With the selected features in our initial projection (**Figure 2a**), we saw that there was a low correlation between the features we wanted to train our model with. There was a 0.48 correlation between the mutation and resistance scoring, suggesting it's an important feature to train our model with, while other features had a low correlation with our target variable. This alone could not conclude which features would be selected to train our model with.

After adding samples from our CRyPTIC dataset and mutation embeddings, we took a look at the 20 most correlated features in the dataset. We saw high correlations mainly between fingerprints, but we had 2 drug embeddings that had high correlation (**Figure 2b**).

3.2.1.2. UMAP Projections

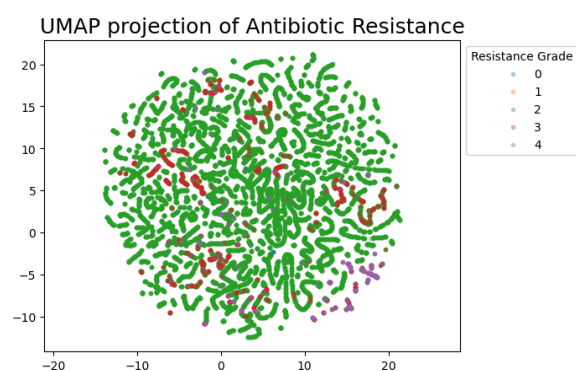


Figure 3a. UMAP projection of all of our features grouped by resistance grade

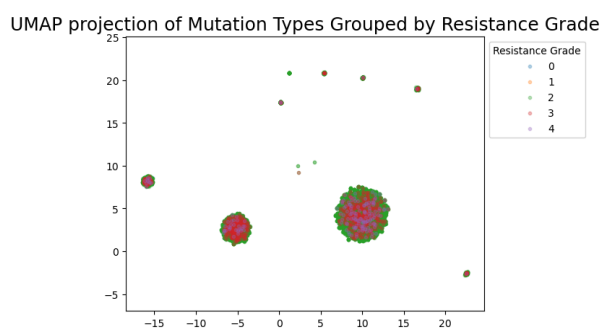


Figure 3b. UMAP projection of mutation types grouped by resistance grade

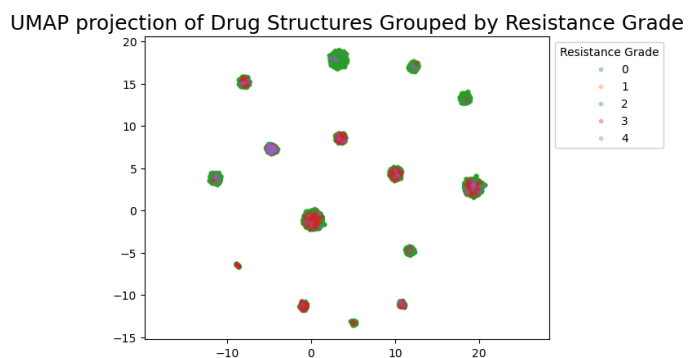


Figure 3c. UMAP projection of antibiotic drug structures grouped by resistance

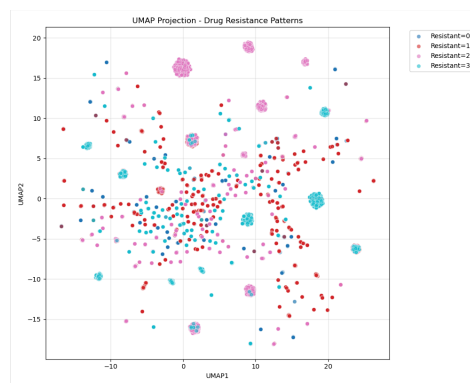


Figure 3d. UMAP projection of resistance grade with CRyPTIC dataset and ChemBERTa embeddings

In our initial UMAP projection of antibiotic resistance grade (**Figure 3a**), we had a very unique clustering that was challenging to interpret. We conducted two other UMAP projections, specifically looking at mutation types (**Figure 3b**) and antibiotic drug structures (**Figure 3c**), to try to get a better understanding. Through our UMAP projections, we saw distinct clusters, but there were overlaps of resistance grade within these clusters.

After adding samples from our CRyPTIC dataset and mutation embeddings, we saw a bit more distinct clustering based on resistance grade (**Figure 3d**). We had a few points that did not cluster at all, but seeing these distinct clusters means there is an underlying structure in our data that our model can learn.

3.2.1.3. PCA

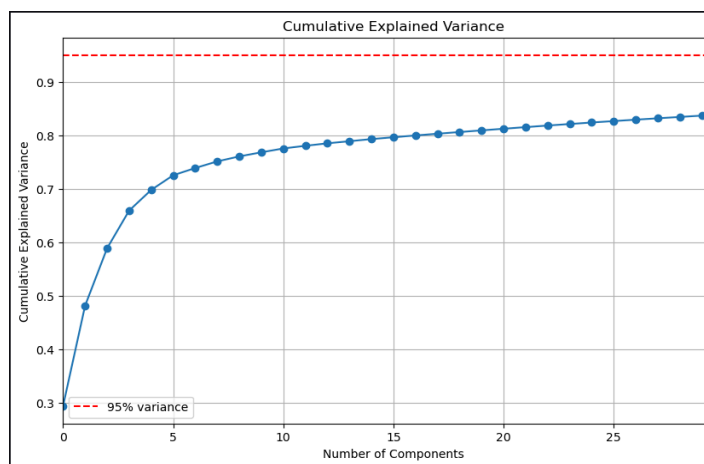


Figure 4a. PCA with all features from the original WHO dataset, with the addition of SMILES strings and Morgan fingerprints from our preprocessing. The cut-off was at 30 components due to the high amount of features after data preprocessing.

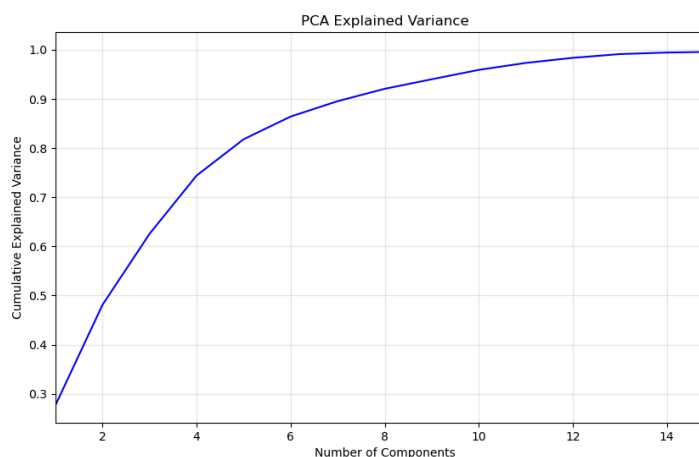
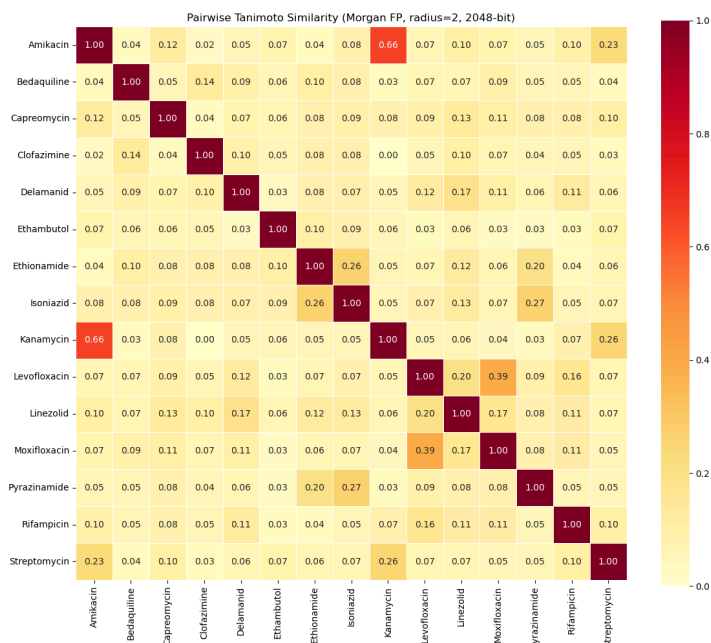


Figure 4b. PCA with CRyPTIC dataset and ChemBERTa embeddings

Our initial PCA (**Figure 4a**) shows that ~70–75% of variance is captured within the first 7–8 components, with minimal gains beyond ~15, indicating that dimensionality can be reduced while preserving key information. After removing some features from the WHO dataset, adding the CRyPTIC dataset and ChemBERTa mutation embeddings to our features, PCA explained variance reached 97% at 12 components (**Figure 4b**).

3.2.1.4. Tanimoto Coefficients

Most drugs show low similarity, though compounds within the same class (e.g., aminoglycosides, fluoroquinolones) cluster together. Few pairs exceed a similarity threshold of 0.7, highlighting the structural diversity of TB drugs (**Figure 5**). We used this information to assist in model development. Withholding drugs with similar features would allow the model to accurately learn the chemical structure.



Drug Pair	Tanimoto	Drug Class
Amikacin – Kanamycin	0.66	Aminoglycosides (30S ribosome)
Levofloxacin – Moxifloxacin	0.39	Fluoroquinolones (DNA gyrase)
Isoniazid – Pyrazinamide	0.27	Small-molecule prodrugs
Isoniazid – Ethionamide	0.26	Mycolic acid synthesis inhibitors
Amikacin – Streptomycin	0.23	Aminoglycosides (30S ribosome)

Figure 5. Tanimoto coefficients calculated (left) with the top 5 highest coefficients (right).

3.3. Logistic Model Evolution

3.3.1. Baseline

We began with a multinomial logistic regression model trained on one-hot encoded mutations and drug names, predicting WHO resistance grade as the target. The encoding produced a high-dimensional sparse feature matrix of roughly 21,000 columns, dominated by mutation indicators. Logistic regression was chosen at this stage for its interpretability and its straightforward handling of high-dimensional sparse inputs, providing a reference point against which subsequent model families and encoding strategies could be compared. The baseline also surfaced an important evaluation consideration: on a heavily imbalanced dataset, aggregate accuracy can be misleading, as strong performance on the majority class can mask poor recall on the minority resistance classes. This motivated our subsequent reliance on macro-averaged metrics and balanced accuracy when evaluating later models.

3.3.2. Drug Structure Similarity Holdout

To assess the generalization to unseen molecular structures, a logistic regression model was retrained again while holding out 3 drugs. Based on the Tanimoto similarity, Amikacin, Kanamycin, and Streptomycin (structurally similar aminoglycosides) were selected and excluded from training to evaluate performance on a chemically similar but unseen drug class.

3.3.3. Boosting Methods

Given that a linear baseline cannot capture nonlinear relationships between mutation profiles and drug identity, we next evaluated gradient-boosted tree ensembles. We initially

applied XGBoost and LightGBM to the same one-hot encoded feature matrix used for the baseline. Both performed substantially worse than logistic regression, which we attributed to the structure of the input: with sparse one-hot indicators across roughly 21,000 mutation and drug columns, individual tree splits gain little information from any given column, and the heavy class imbalance further pushes the trees toward over-correcting on minority classes. We hypothesized that the bottleneck was representational rather than model-capacity-related.

We then evaluated CatBoost, which replaces sparse one-hot columns with ordered target statistics: each categorical value is represented as a single dense column encoding its running label-mean rather than thousands of sparse indicators. This addresses the encoding bottleneck directly, without changing model capacity. We further trained an extended configuration ("CatBoost Extended") that added gene identity and the Morgan fingerprint of each drug as additional features, providing substructural chemical information beyond drug name alone and supporting the project's broader goal of structure-aware modeling.

To understand which features the extended model relied on, we computed feature importance scores. The relative weight that CatBoost placed on chemical substructure versus drug identity served as a diagnostic for whether the model was learning chemistry-driven patterns — the behavior we would want from a model intended to generalize to novel compounds — and motivated the drug-holdout stress test described in Section 3.3.2.

3.4. ANN Model

A simple artificial neural network was implemented using a multi-layer perceptron to capture potential nonlinear relationships between mutation, gene, and chemical fingerprint features. The model consists of two hidden layers, with 64 and 32 neurons respectively, each using ReLU activation. This structure allows the model to learn interactions between features that linear models like logistic regression cannot capture. This model was small enough to avoid overfitting, large enough to capture interactions, and progressively compresses information.

Two hidden layers with 64 and 32 neurons were used to give the model enough capacity to learn interactions between features while keeping it small enough to avoid overfitting. The decreasing size helps the network gradually compress information into more useful representations. ReLU activation introduces nonlinearity, allowing the model to capture complex relationships between mutation, gene, and chemical features, and it is efficient and widely used in practice. 40 epochs were chosen because they were enough for the model to converge.

4. Evaluation

4.1. Model Results with the Target as WHO Resistance Grades

4.1.1. Logistic Regression Model - Random Split (Target: Who Grade)

A logistic regression model was trained as a baseline model to predict WHO resistance grades using a traditional random train/test split. The model achieved strong performance, with a balanced accuracy score of 0.918 (**Figure 6**). However, this evaluation is likely overly optimistic due to data leakage from the random split; identical mutations and chemical structures may be

present across both train and test data. This allows the model to see similar mutations and chemical structures during training, leading to an inflated performance.

Despite the potential data leakage, some ambiguity between the grades still remains. For example, the model still misclassified “Not associated with resistance - interim” as “Not associated with resistance” in 21% of cases. The confusion indicates there is still a subtle distinction between classes, which mainly differ in confidence of evidence rather than biological differences, and highlights the challenge of learning from partially uncertain or evolving clinical annotations.

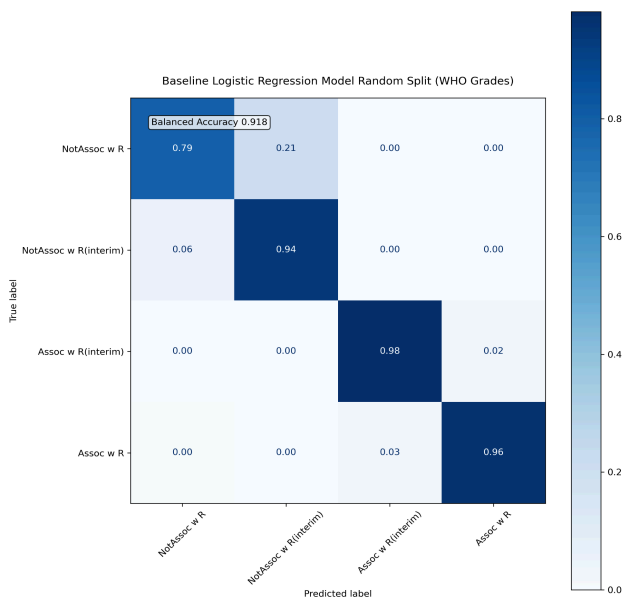


Figure 6. Confusion matrix of baseline logistic regression model using a random train split with a balanced accuracy score of 0.92.

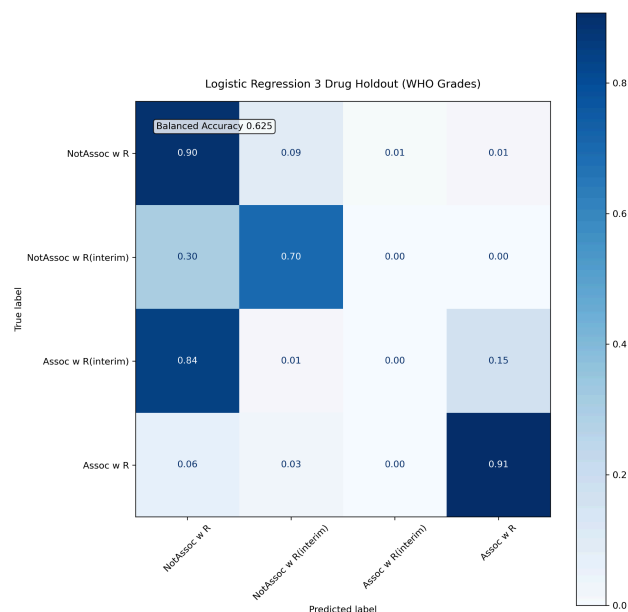
4.1.2. Logistic Regression Model - Drug Holdout (Target: Who Grade)

Model performance decreased significantly with a balance accuracy of 0.62, indicating limited ability to generalize to unseen drugs (**Figure 7a**). While the model was able to distinguish broad resistant classes (class 2-3) from non-resistant (class 0-1), it struggled to classify intermediate and uncertain classes. These misclassifications suggest the model collapses uncertain classes into more confident resistance classes (**Figure 7b**).

Even within aminoglycosides, resistance mechanisms differ, mutation effects differ, and MIC/PPV mapping differs, making it difficult for the model to capture resistance based solely on mutation and structural features.

Additionally, another limitation of logistic regression is that it assumes linear relationships. Given that our problem is highly dimensional and likely nonlinear between mutation profiles and drug representations, this limitation hurts performance. Even with

advanced molecular representations (ChemBERTa embeddings + Morgan fingerprints), predicting resistance for unseen drugs remains challenging.



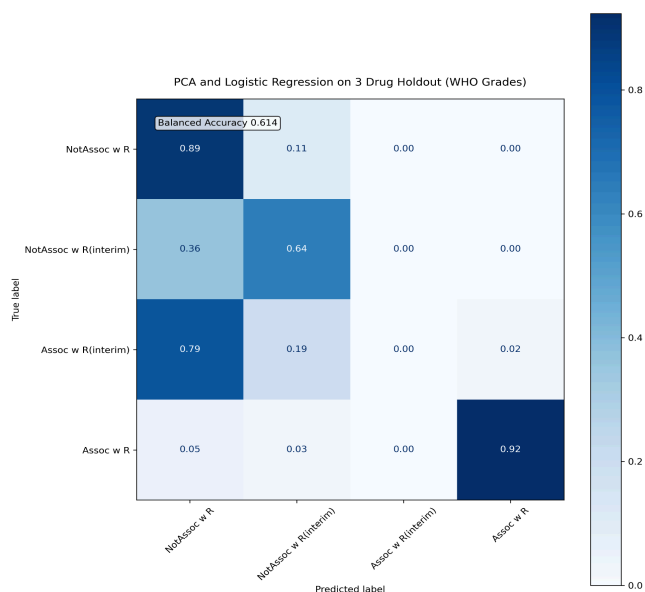
Classification Report for Logistic Regression on Holdout Test Set:				
	precision	recall	f1-score	support
0	0.34	0.90	0.49	353
1	0.85	0.70	0.77	501
2	0.00	0.00	0.00	486
3	0.91	0.91	0.91	809
accuracy			0.65	2149
macro avg	0.52	0.62	0.54	2149
weighted avg	0.60	0.65	0.60	2149

Figure 7b. Classification report for logistic regression model tested on a subset of 3 aminoglycoside drugs. Associated with resistance (interim) class 2 completely collapsed.

Figure 7a. Confusion matrix of logistic regression model tested on a subset of 3 aminoglycoside drugs with a balanced accuracy score of 0.62.

4.1.3. PCA + Logistic Regression - Drug Holdout (Target: WHO Grade)

Applying PCA before this model did not improve performance, giving a balanced accuracy of 0.61 (**Figure 8a**). Similar to the previous logistic regression model, the intermediate resistance classes collapse, suggesting the most informative features for predicting resistance are not captured. PCA may also cause important localized mutations or structural drug features to be combined into global components. As a result, reducing dimensionality does not address our underlying limitation of logistic regression to model nonlinear relationships between mutations and drug structure.



	precision	recall	f1-score	support
0	0.34	0.89	0.50	353
1	0.67	0.64	0.65	501
2	0.00	0.00	0.00	486
3	0.99	0.92	0.96	809
accuracy			0.64	2149
macro avg	0.50	0.61	0.53	2149
weighted avg	0.59	0.64	0.59	2149

Figure 8b. Classification report after running PCA before the logistic regression model was tested on a subset of 3 aminoglycoside drugs. Associated with resistance (interim) class 2 completely collapsed.

Figure 8a. Confusion matrix after running PCA before logistic regression model tested on a subset of 3 aminoglycoside drugs with a balanced accuracy score of 0.61.

4.1.4. ANN - Drug Holdout (Target: WHO Grade)

To further improve on generalization, an artificial neural network (ANN) was trained under the same drug holdout test set. The ANN model achieved a higher performance with a balanced accuracy of 0.85 compared to logistic regression at only 0.62, indicating a great improvement in learning unseen drug-mutation relationships (**Figure 9**). The confusion matrix also supports improvements across all classes, and unlike logistic regression, the ANN was especially effective at separating the intermediate resistance classes.

The improvement is likely due to ANN's ability to capture nonlinear relationships, unlike logistic regression. Given the complexity of the resistance mechanisms and drug-mutation relationships, the nonlinear effects were essential for accurate prediction of WHO grades, suggesting that changing to a more flexible model can significantly improve generalization to unseen drugs.

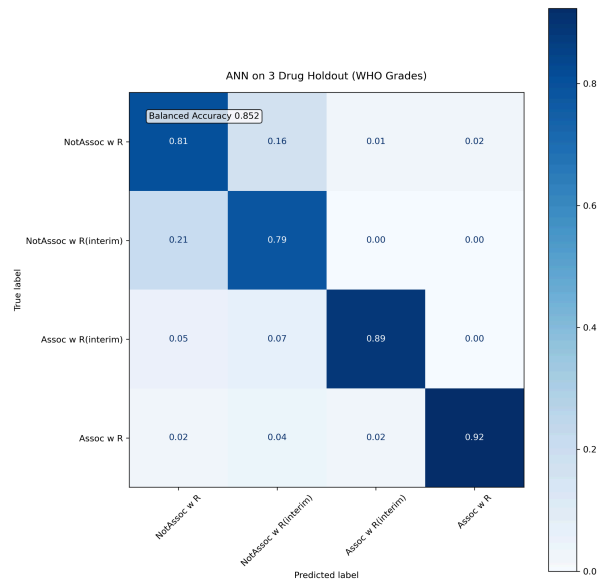


Figure 9. Confusion matrix after running ANN model tested on a subset of 3 aminoglycoside drugs with a balanced accuracy score of 0.85.

4.1.5. Boosting Methods (Target: Who Grade)

When we swapped in XGBoost and LightGBM on the same one-hot features, both performed substantially worse: XGBoost at 42% accuracy, LightGBM at 32%. For CatBoost, we change only how the categorical columns are encoded. Balanced accuracy jumps from XGBoost's 36% to CatBoost's 62% — almost doubling. CatBoost Extended adds gene identity and the Morgan fingerprint of each drug. Balanced accuracy rises again to 68% — the best of all five models, beating logistic regression by 12% on the metric (**Figure 10**).

Model	Accuracy	Balanced Acc	Macro F1	Weighted F1
Logistic Regression	0.800	0.562	0.452	0.824
LightGBM	0.320	0.421	0.202	0.361
XGBoost	0.416	0.361	0.272	0.454
CatBoost	0.624	0.617	0.382	0.739
CatBoost Extended	0.699	0.680	0.431	0.775

Figure 10. Results of boosting methods when applied to our model.

4.2. Pivoting Target from WHO Resistance Grades to Positive Predicted Value

While the WHO resistance grade provides a clinically interpretable classification, they are discrete, drug-specific, and heavily dependent on expert opinion. This introduced ambiguity, especially the “interim” classes, which we discovered in our first models, and also limits generalizability to new drugs.

To address these limitations, we decided to shift to modeling positive predictive values (PPV), a continuous and data-driven measurement of antibiotic resistance. PPV measures the probability that a mutation will cause resistance to a drug, which is more universal and quantitative for representing the effects of mutations and antibiotic resistance.

Additionally, given the limited number of drugs in the dataset (15 TB drugs) and the ambiguity of WHO grades, there is still a risk of a model learning drug-specific patterns rather than generalizing mutation effects. By focusing on PPV, we aimed to build models that can more accurately generalize to novel mutations and capture biological relationships independent of a specific drug. While the current dataset is still specific to TB, the PPV-based framework could be extended to other bacterial pathogens if similar drug-mutation resistance datasets are available.

4.2.1. Random Forest Regression Random Split (Target: PPV)

A Random Forest regressor was trained as a baseline model to predict continuous PPV values using a random train/test split. The model performed well on the test set (MAE=0.043 and $R^2=0.82$), indicating there is a strong correlation between predicted vs actual PPV values (**Figure 11**). Compared to the train set, we see the model likely overestimated performance. Similar to the random split on the logistic regression model, there is data leakage, and identical mutations and drug structures were present in both sets of data, allowing the model to memorize without effectively learning patterns, leading to an inflated performance score (train $R^2 = 0.97$).

Despite the overfitting, the model is still able to capture strong relationships between features, and increased variance on the lower PPV values suggests a lower probability of resistance is still more challenging to capture.

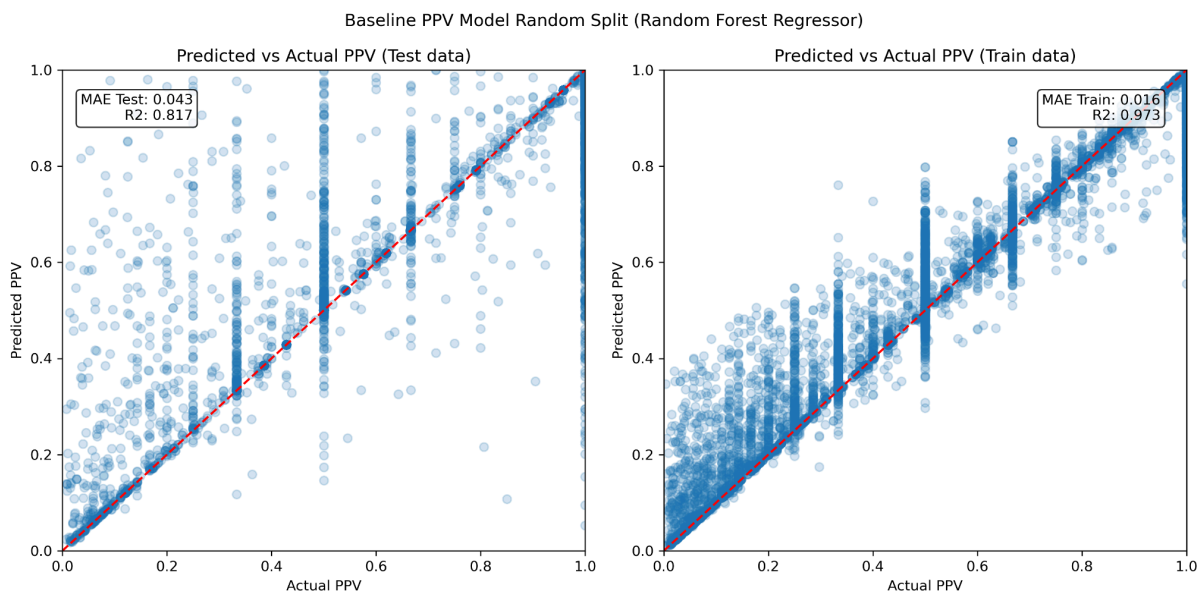


Figure 11. Predicted PPV scatter plots of test data vs training data using random forest regression as a baseline model with a random train split.

4.2.2. Random Forest Regression GroupKFold (Target: PPV)

To examine generalizability to novel unseen mutations, a Random Forest regressor was trained using GroupKFold to prevent data leakage and test the robustness of our model. Five groups of 2500 unseen mutations were hidden from the model during training to evaluate model performance. Under this more rigorous model, the model performance decreased (MAE=0.06 and $R^2 = 0.7$). Despite the drop, the linear relationship between predicted and actual PPV scores indicated the model was able to capture generalizable patterns within mutation features and drug structure (**Figure 12**). Using a Random Forest regressor showed that models can effectively learn mutation information linked to resistance relationships even in the presence of noisy and limited data.

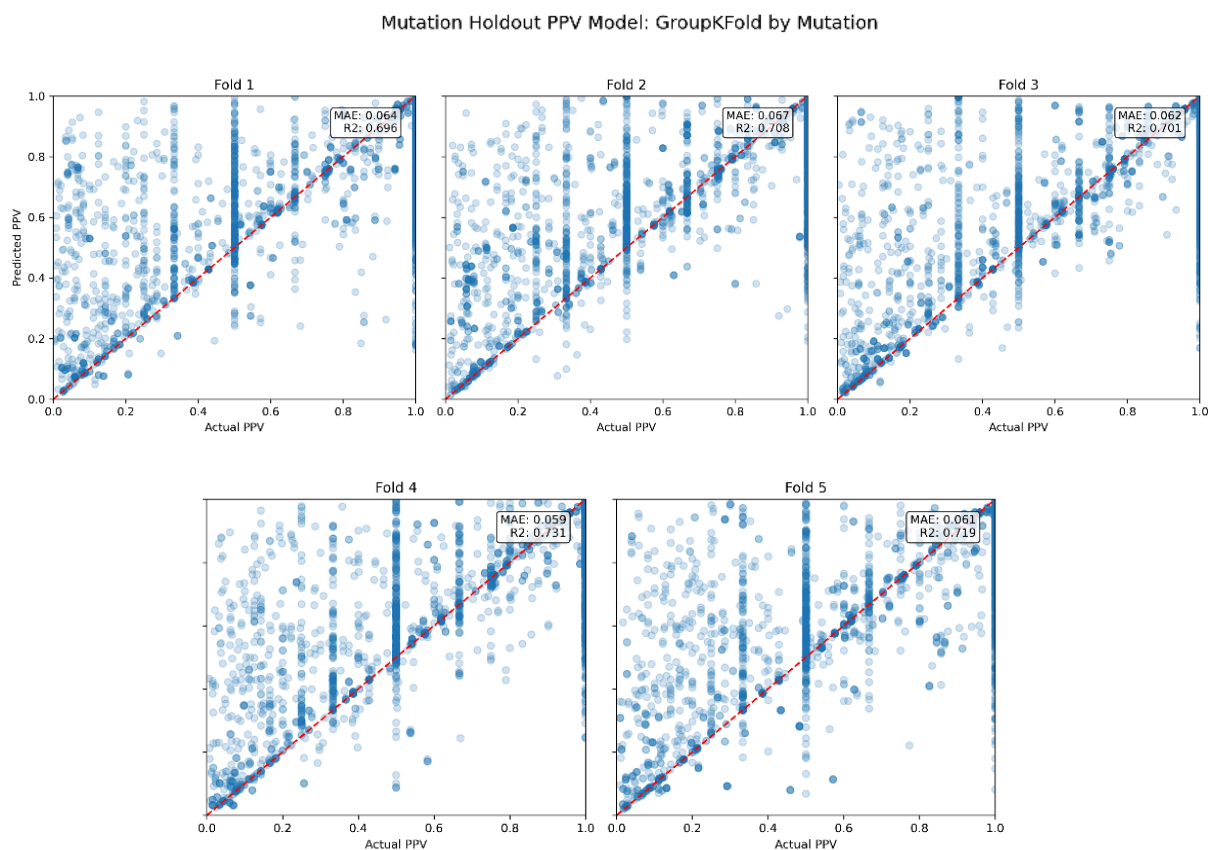


Figure 12. Predicted PPV scatter plots using random forest regression tested on five sets of 2500 hidden mutations using GroupKFold.

4.2.3. ANN, Logistic Regression, and Random Forest Classifiers (Target: PPV Bins)

Due to the limited and noisy data, PPV values were categorized into risk bins and then modeled using ANN, logistic regression, and Random Forest classifiers. Both logistic regression

and ANN achieved reasonable performances (balanced accuracies of 0.79 and 0.78, respectively), the Random Forest model consistently outperformed with a balanced accuracy of 0.82, with the least number of misclassifications (**Figure 13**).

While ANN models have more flexibility, the PPV bins produced more separable classes, allowing models like logistic regression to perform comparably well. However, the Random Forest model was still better at handling nonlinear interactions as well as noisy targets like PPV, leading to a more robust performance.

Based on these results, the Random Forest classifier on PPV bins was selected as the final model, providing the best accuracy and handling unseen mutations. This suggests that tree-based models are best suited for modeling mutation-driven resistance patterns in high-dimensional biological data.

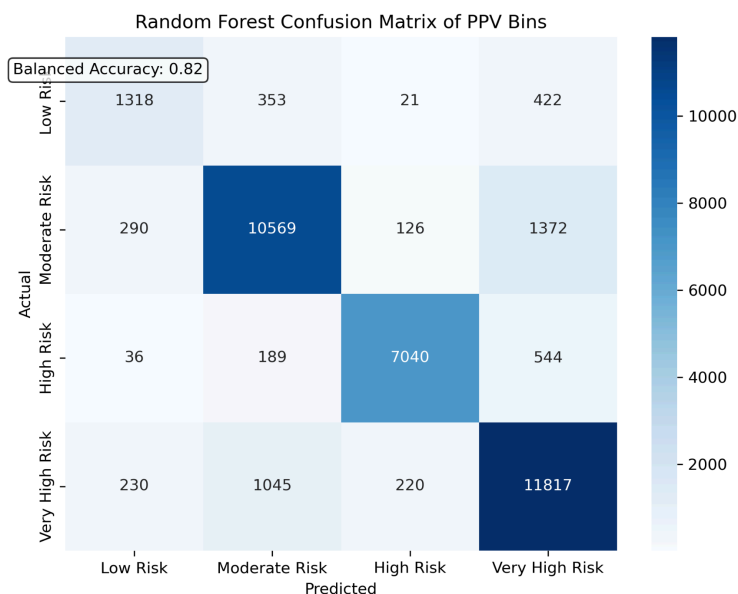


Figure 13. Summary confusion matrix of PPV bins using random forest regression tested on five sets of 2500 hidden mutations using GroupKFold.

4.2.4. Boosting Methods (Target: PPV Bins)

To evaluate if the Random Forest baseline can be improved using boosting methods, we trained and evaluated 4 additional models: AdaBoost, XGBoost, LightGBM, and CatBoost.

The gradient boosting models (XGBoost, LightGBM, and CatBoost) outperformed AdaBoost. XGBoost achieved the highest balanced accuracy at 0.822 (**Figure 14b**). LightGBM and CatBoost ranked second with 0.821 (**Figure 14c** and **Figure 14d**, respectively). And AdaBoost had the lowest balanced accuracy of 0.711 (**Figure 14a**). AdaBoost may have repeatedly increased the weight of misclassified samples, which can cause the model to overemphasize noisy examples near PPV-bin boundaries, thus the underperformance. Moreover,

the similar performances between gradient-boosted tree models and the Random Forest classifier suggest that the choice among these tree-based nonlinear models had little effect on this dataset.

The confusion matrices revealed a consistent failure mode across all boosting models: Low Risk samples were sometimes misclassified as Very High Risk. Even the best-performing models misclassified a substantial number of true Low Risk samples into the Very High Risk class. This suggests that mutation alone might not be sufficient for classifying binned PPVs. If the model’s features do not adequately capture drug-specific differences, it may overgeneralize and misclassify low-risk mutation–drug pairs. In contrast, High Risk and Very High Risk samples had better classification accuracy. This suggests that strong resistance-associated patterns were easier for the models to identify.

In summary, boosting methods did not meaningfully improve the performance of classifying binned PPVs. This suggests that the bottleneck is likely related to data rather than model capacity.

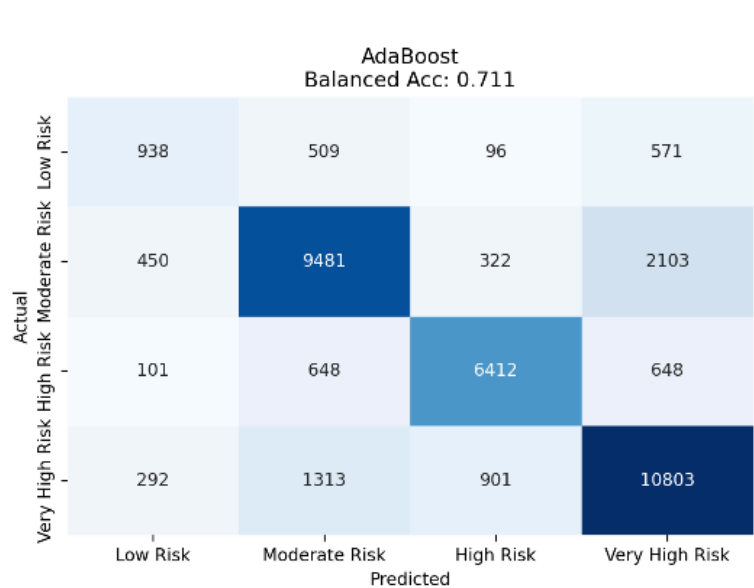


Figure 14a. Summary confusion matrix of PPV bins using AdaBoost, tested on five sets of 2500 hidden mutations using GroupKFold.

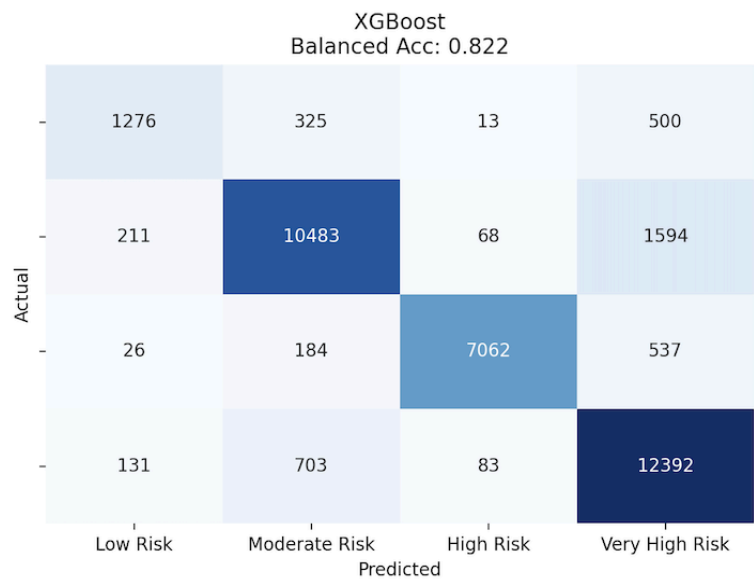


Figure 14b. Summary confusion matrix of PPV bins using XGBoost, tested on five sets of 2500 hidden mutations using GroupKFold.

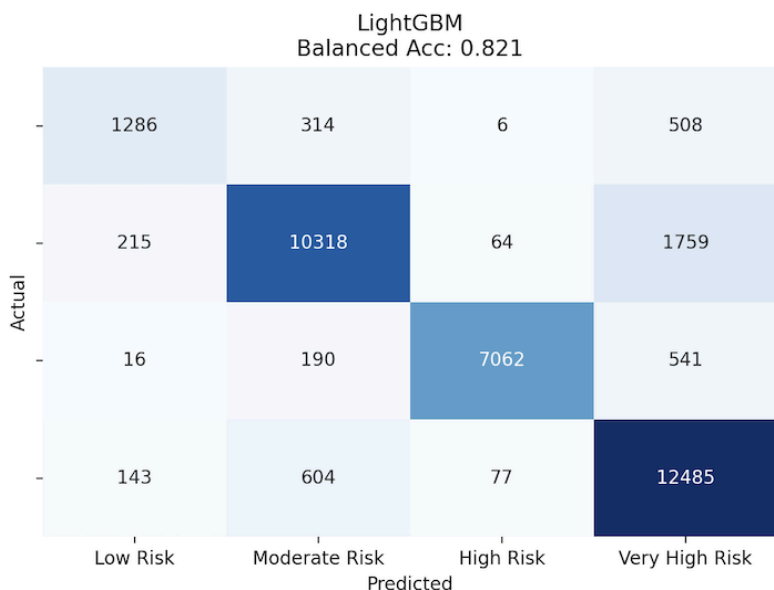


Figure 14c. Summary confusion matrix of PPV bins using LightGBM, tested on five sets of 2500 hidden mutations using GroupKFold.

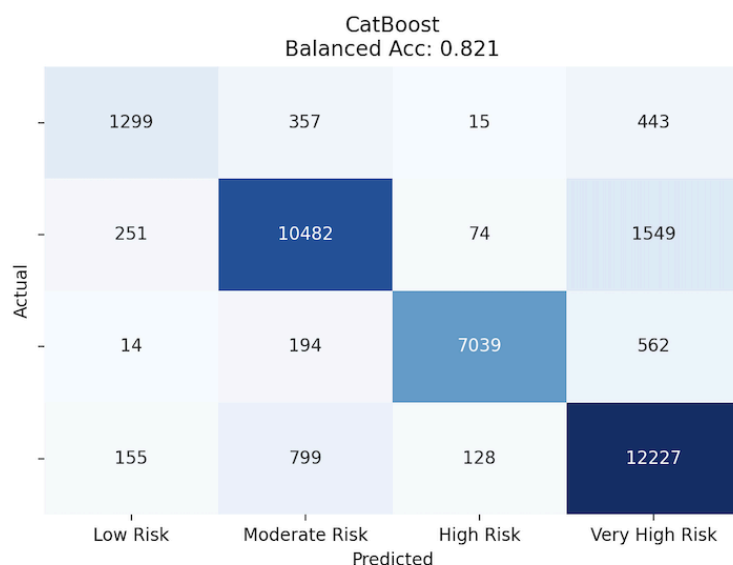


Figure 14d. Summary confusion matrix of PPV bins using CatBoost, tested on five sets of 2500 hidden mutations using GroupKFold.

5. Discussion and Conclusion

5.1. Interpret results

We observed a significant improvement in performance after switching the prediction target from WHO resistance grade to PPV bins. This supported our initial hypothesis that the WHO resistance grading framework was too inconsistent for our models to effectively generalize the mutation-associated resistance effects of TB drugs. Specifically, the WHO “interim” classes introduced uncertainty and misclassifications between resistant and nonresistant classes, making it difficult to learn decision boundaries.

When using PPV bins as the target, both Random Forest and XGBoost achieved strong performance scores across unseen mutation groups with balanced accuracy scores of around 82% or greater. However, XGBoost was selected as the best performing model with improved sensitivity when predicting high-risk resistance categories and fewer false negatives than Random Forest. False positives are preferable because false negative predictions to resistance could result in a more conservative but effective treatment, while false positives could lead to ineffective treatment.

5.2. Improvements

One improvement that was briefly tested was training our own embeddings for the gene and position combination, as well as a smaller trained embedding for the drugs. This would reduce the dimensions of our dataset further, reducing the sparsity and providing a more meaningful signal, since the same position can be associated with a drastically different protein function on different genes.

5.3. Conclusion

In this project, multiple machine learning approaches were evaluated to predict antimicrobial resistance using genomic mutation features and molecular drug representations. Initial models trained on the WHO resistance grades highlighted strong class imbalances present in the raw dataset and the ambiguity of the WHO grading scheme, resulting in weak models that struggled to distinguish between biologically meaningful predictions. As a result, we pivoted towards feature engineering to improve the quality, relevance, and robustness of our initial data and changed our target to PPV resistance measurement.

Overall, tree-based methods, particularly XGBoost, achieved a strong performance for PPV-bin classification and demonstrated the ability to learn and generalize to unseen mutations. These results highlighted the importance of feature selection and feature engineering to improve antimicrobial resistance predictions. By combining custom-encoded genomic mutation features and position data with traditional Morgan fingerprints and deep ChemBERTa embeddings, the models were able to capture information about genomic mutation features and drug structure. This framework allowed us to create a model that supports future development of data-driven antimicrobial resistance prediction and aids in more informed patient treatment decisions of *Mycobacterium tuberculosis*.

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